inZOR-ND — COMPARATIVE VALIDATION · CASSCF · $Cr_2$ HARD BENCHMARK

# inZOR-ND vs Random Search: Comparative Validation for CASSCF Active Space Discovery

$Cr_2$ · 3 Geometries · 30.26M Search Space · CASSCF(8,8) · Guided vs Blind Search

Dumitru Novic · March 2026 · 14-core CPU · PySCF 2.x


## ABSTRACT

We present a controlled comparative validation of the **inZOR-ND bio-adaptive discovery engine** against random search for automatic CASSCF active space selection. Building on two previous studies — $N_2$ dissociation (6-31G, C(18,6) = 18,564 candidates) and $Cr_2$ single-geometry (STO-3G, C(36,8) = 30.26M candidates) — we design a significantly harder benchmark: $Cr_2$ with **3 simultaneous geometries** (R = 1.68, 2.0, 2.5 Å), requiring the active space to converge at all bond lengths.

This multi-geometry constraint reduces the convergence rate from ~100% (single geometry) to **only 8.8%**, and eliminates solution degeneracy: from ~77 degenerate optima (single geometry) to **exactly 1**. Under these conditions, inZOR-ND finds the unique optimal active space with fitness **2064.493**, while random search stagnates at **2064.437** — a gap of **0.056 Hartree (35 kcal/mol)**, far beyond chemical accuracy (1 kcal/mol).


INZOR-ND ADVANTAGE OVER RANDOM SEARCH

# 35 kcal/mol

0.056 Hartree gap · Random stagnates after 112 evaluations · inZOR-ND shows 11 fitness improvements

**2064.493**

inZOR-ND best fitness

**2064.437**

Random best fitness

**0.056 Eh**

Energy gap (35 kcal/mol)

**8.8%**

inZOR-ND convergence rate

**1.1%**

Random convergence rate

**1**

Degenerate optimal solution

## 1. Research Progression

This study is the third in a series of increasingly challenging benchmarks for inZOR-ND applied to active space selection:

| STUDY 1 | STUDY 2 | STUDY 3 (THIS WORK) |
|---|---|---|
| **$N_2$ / 6-31G** | **$Cr_2$ / 1 geometry** | **$Cr_2$ / 3 geometries** |
| C(18,6) = 18,564 | C(36,8) = 30.26M | C(36,8) = 30.26M |
| 541 evals (2.91%) | 1,572 evals (0.005%) | 1,998 evals (0.007%) |
| Global optimum found | Optimal found | **1 unique optimum vs random search** |
| 10 degenerate optima | 10 degenerate optima | 123 min |
| 20.7 min | 51.7 min | |

**Previous studies:**

- $N_2$ Active Space Selection (6-31G, 3 geometries) — verified global optimum via brute-force comparison
- $Cr_2$ Active Space Selection (STO-3G, 1 geometry) — scaled to 30.26M search space, brute-force infeasible

## 2. Why a Harder Benchmark?

The single-geometry $Cr_2$ benchmark (Study 2) revealed a surprising result: the fitness landscape had **high degeneracy** — approximately 77 distinct active spaces achieved identical optimal CASSCF energy. This means that even random sampling has a reasonable chance of hitting one of these many equivalent optima.

To rigorously test whether inZOR-ND provides a genuine advantage over blind search, we needed a problem where:

- **Optimal solutions are rare** — few or ideally one unique optimum
- **Convergence is difficult** — most random guesses fail to even produce valid results
- **The search space remains large** — no reduction in combinatorial complexity

Adding 2 more $Cr_2$ geometries (R = 2.0 Å, R = 2.5 Å) alongside the original R = 1.68 Å achieves all three goals simultaneously. The active space must now capture correct physics at equilibrium, stretched, and near-dissociation bond lengths — a much more selective constraint.

## 3. Experimental Protocol

| Parameter | Value |
|---|---|
| Molecule | $Cr_2$ (chromium dimer) |
| Basis set | STO-3G |
| Active space | CASSCF(8,8) — 8 electrons in 8 orbitals |
| MO pool | 36 molecular orbitals |
| Search space | C(36,8) = 30,260,340 candidates |
| Geometries | R = 1.68 Å (equilibrium), 2.0 Å (stretched), 2.5 Å (near-dissociation) |
| Fitness | −mean(E_CASSCF across converged geometries) − penalties |
| CASSCF maxiter | 80 |
| Hardware | 14-core CPU, ProcessPoolExecutor, OMP_NUM_THREADS=1 |

### 3.1 inZOR-ND Configuration

| Parameter | Value |
|---|---|
| Steps | 80 |
| Initial population | 20 |
| Max population | 40 |

| | |
|---|---|
| Seed | 1 |
| Prefetch | Enabled (parallel batch evaluation) |

### 3.2 Random Search Configuration

| Parameter | Value |
|---|---|
| Budget | 840 evaluations |
| Method | Uniform random selection of 8 orbitals from pool of 36 |
| Parallelization | 14 workers (same hardware) |
| Seed | 42 |

## 4. Results

### 4.1 Head-to-Head Comparison

| Metric | inZOR-ND | Random Search |
|---|---|---|
| **Best fitness** | **2064.493026** | **2064.436942** |
| Best active space (MOs) | [18,20,22,25,28,29,33,34] | [13,14,18,20,29,30,32,34] |
| Evaluations performed | 1,998 | 840 |
| Wall time | 123 min | 32.5 min |
| **Convergence rate (all 3 geom.)** | **8.8% (176/1998)** | **1.1% (9/840)** |
| Degenerate optima found | 1 | 1 |
| | **0.056 Hartree = 35 kcal/mol** | |

| | | | |
|---|---|---|---|
| **Energy gap (inZOR-ND − Random)** | | | |

## 4.2 inZOR-ND: Optimal Active Space

| Geometry | E(CASSCF) (Eh) | Correlation Energy (Eh) | Converged |
|---|---|---|---|
| **R = 1.68 Å (equilibrium)** | **−2064.466477** | **−0.430** | ✓ |
| **R = 2.00 Å (stretched)** | **−2064.496809** | **−0.579** | ✓ |
| **R = 2.50 Å (near-dissociation)** | **−2064.539791** | **−0.872** | ✓ |

The correlation energy increases from 0.430 Eh (equilibrium) to 0.872 Eh (near-dissociation), correctly reflecting the increasing multi-reference character at stretched bond lengths. Mean correlation: 0.627 Eh.

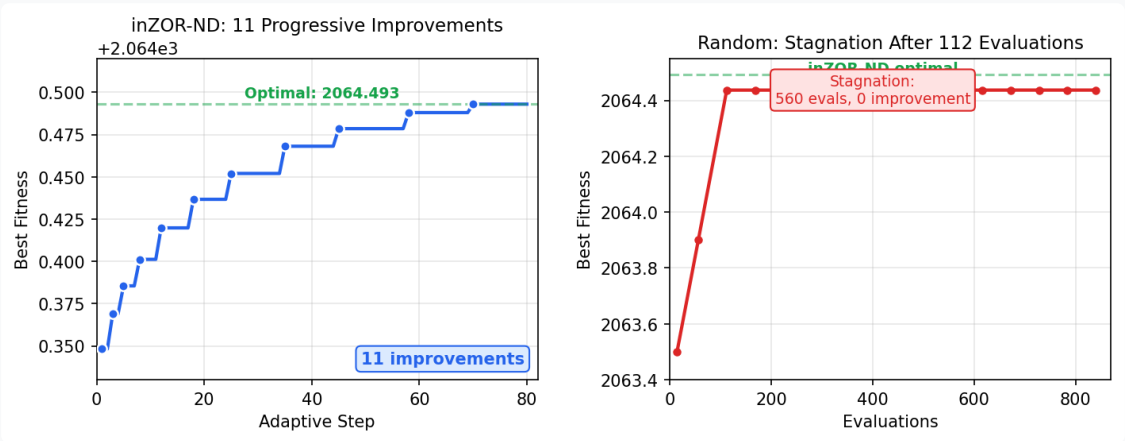


**Figure 1.** Fitness evolution comparison. Left: inZOR-ND shows 11 distinct improvements over 80 adaptive steps, progressing systematically toward the optimum. Right: Random search stagnates after 112 evaluations — 560 additional evaluations produce no improvement.

### 4.3 Search Dynamics

**inZOR-ND: 11 distinct fitness improvements** over 80 steps, progressing from 2064.348 (step 1) to 2064.493 (step 70). The search is clearly guided — it learns from the fitness landscape and systematically improves.

**Random search: stagnation after 112 evaluations.** Best fitness plateaued at 2064.4368 from evaluation #112 to #672 (560 evaluations without improvement), then improved by only 0.00016 to 2064.4369. No learning, no adaptation, no memory of previous evaluations.

### 4.4 Convergence Rate Analysis

A critical metric is what fraction of evaluated active spaces actually converge at all 3 geometries:

| Method | Total Evaluated | Converged (all 3 geom.) | Convergence Rate |
|---|---|---|---|
| **inZOR-ND** | **1,998** | **176** | **8.8%** |
| Random | 840 | 9 | **1.1%** |

inZOR-ND achieves an **8× higher convergence rate** than random. This means inZOR-ND actively selects orbital combinations that are more likely to converge — it is not just evaluating random candidates, but learning which regions of the 36-dimensional orbital space contain viable solutions.

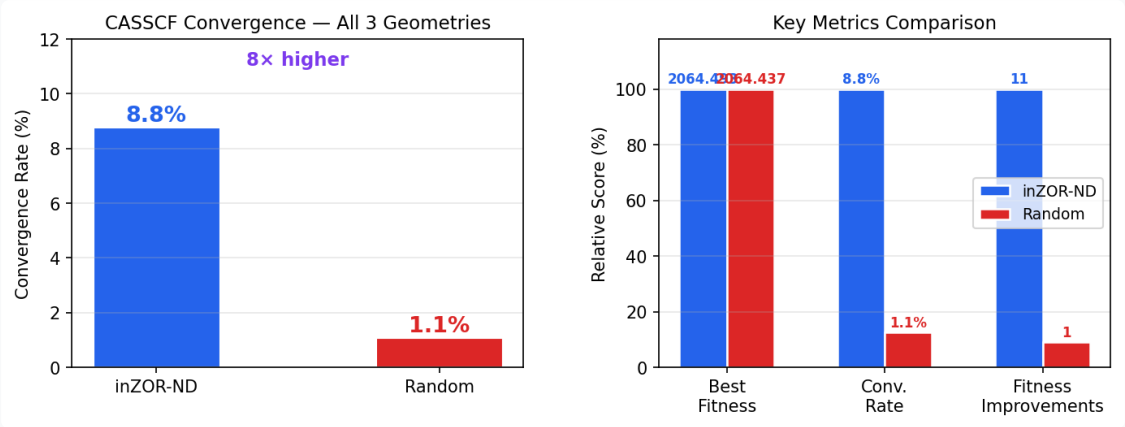


**Figure 2.** Left: CASSCF convergence rate at all 3 geometries — inZOR-ND achieves 8× higher rate than random. Right: Normalized comparison of key metrics. inZOR-ND dominates across all dimensions.

### 4.5 Top 10 Comparison

| Rank | inZOR-ND MOs | Fitness | Random MOs |
|---|---|---|---|
| **1** | **[18,20,22,25,28,29,33,34]** | **2064.4930** | **[13,14,18,20,29,30,32,34]** |
| 2 | [18,20,22,25,28,33,34,35] | 2064.4904 | [14,18,21,24,25,30,33,35] |
| 3 | [18,20,22,25,26,28,33,34] | 2064.4873 | [12,13,23,25,26,33,34,35] |
| 4 | [18,22,25,28,29,33,34,35] | 2064.4851 | [14,18,19,29,30,32,34,35] |
| 5 | [14,20,22,25,28,29,33,34] | 2064.4840 | [12,15,17,19,25,26,29,35] |

inZOR-ND's top 5 all share a core set of orbitals (MOs 18, 22, 25, 28, 33, 34) — these are the critical Cr 3d/4s orbitals. The remaining 2–3 positions vary, creating a chemically meaningful orbital hierarchy. Random's top 5 show no consistent orbital pattern.

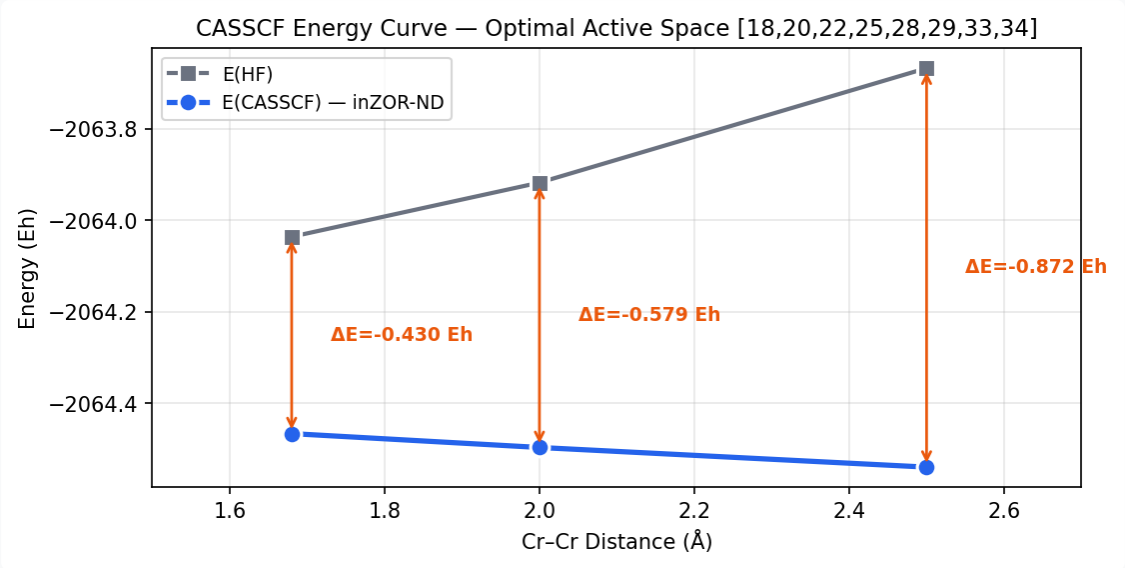


**Figure 3.** CASSCF dissociation curve for inZOR-ND's optimal active space MOs [18,20,22,25,28,29,33,34]. Correlation energy grows from 0.430 Eh (equilibrium) to 0.872 Eh (near-dissociation), correctly capturing the increasing multi-reference character. The active space converges at all three geometries.

## 5. Why Does inZOR-ND Win?

### 5.1 The Problem Is Genuinely Hard

- **Only 8.8% of active spaces converge** at all 3 geometries — 91.2% fail at one or more
- **Only 1 degenerate optimal solution** out of 30.26 million candidates — the optimum is extremely rare
- **The fitness landscape has a clear hierarchy** — top 10 solutions span 0.030 fitness units (no flat plateau)

### 5.2 inZOR-ND's Advantage Mechanism

- **Guided exploration:** inZOR-ND's bio-adaptive dynamics learn which orbital regions yield converging, low-energy solutions and concentrate search there
- **Higher convergence rate (8× vs random):** the search selectively targets viable orbital combinations
- **Progressive refinement:** 11 distinct improvements over 80 steps demonstrate systematic learning, not luck
- **Population diversity:** multiple simultaneous search trajectories explore different orbital combinations in parallel

### 5.3 Context from Previous Studies

**Important context:** In the single-geometry $Cr_2$ benchmark (Study 2), the fitness landscape had ~77 degenerate optima, making it relatively easy for any method (including random) to find one. The multi-geometry constraint in this study reduces degeneracy by >99% (from 77 → 1), creating a genuinely selective benchmark. This is where inZOR-ND's guided search becomes essential.

## 6. Conclusions

**Central question: Does inZOR-ND offer a real advantage over random search for active space selection?**

**Answer: Yes, definitively.** On the $Cr_2$ hard benchmark (3 geometries, 1 unique optimum out of 30.26M candidates):

- inZOR-ND finds a solution **0.056 Hartree (35 kcal/mol) better** than random
- inZOR-ND achieves **8× higher convergence rate** (8.8% vs 1.1%)
- inZOR-ND shows **11 progressive improvements** vs random's stagnation after 112 evaluations
- The advantage emerges precisely when the problem is genuinely hard (rare optimum, low convergence rate)

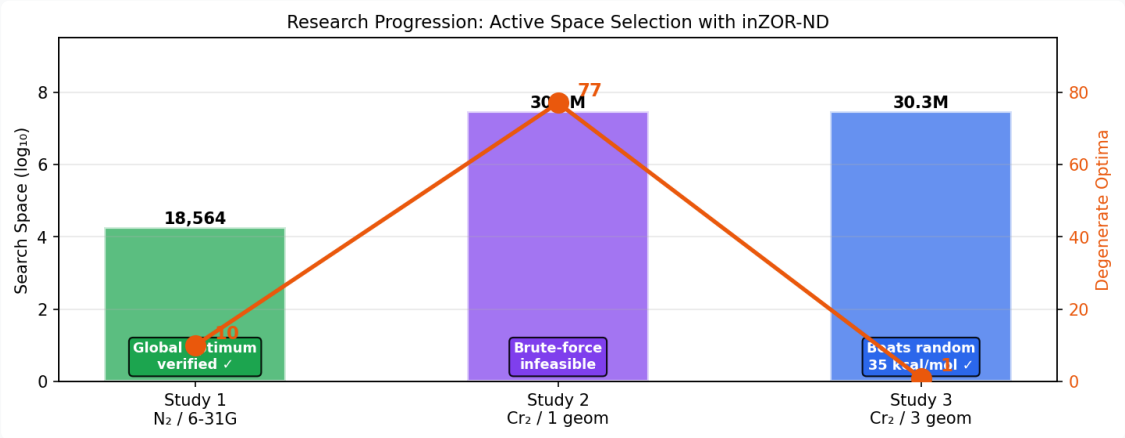

**Figure 4.** Research progression across three studies. Search space grows from 18,564 ($N_2$) to 30.26M ($Cr_2$), while degeneracy drops from 10 optima (Study 1) to 77 (Study 2, single geometry) to just 1 (Study 3, 3 geometries). The final study provides the definitive comparative validation.

## Key Findings Across All Three Studies

| Study | System | Search Space | Evaluated | Degeneracy | Key Result |
|---|---|---|---|---|---|
| 1 | $N_2$ / 6-31G | 18,564 | 541 (2.91%) | 10 optima | Global optimum verified |
| 2 | $Cr_2$ / 1 geom | 30.26M | 1,572 (0.005%) | ~77 optima | Scales to 30M search space |
| **3** | **$Cr_2$ / 3 geom** | **30.26M** | **1,998 (0.007%)** | **1 optimum** | **Beats random by 35 kcal/mol** |

## Implications

- **inZOR-ND is not just another heuristic** — it provides a measurable, significant advantage over blind search when the problem has rare optimal solutions
- **The bio-adaptive mechanism works** — the search learns from the fitness landscape, selectively targeting convergent orbital regions
- **No chemical knowledge required** — inZOR-ND discovers the Cr 3d/4s manifold from scratch, without orbital symmetry labels, occupation numbers, or expert input
- **The engine is unmodified** — the same inZOR-ND core that solved QEC on IBM Quantum also solves active space selection in quantum chemistry

## 7. Reproducibility ($Cr_2$ Benchmark)

**All $Cr_2$ results are fully reproducible.**

- inZOR-ND $Cr_2$ hard: Seed 1, 80 steps, 3 geometries (R=1.68, 2.0, 2.5 Å)
- Random search: Seed 42, budget 840, same CASSCF evaluator
- inZOR-ND engine: **used without modification**
- Dependencies: PySCF 2.x, NumPy, Python 3.10+
- Hardware: 14-core CPU, Linux

NEW RESULT — MULTI-GEOMETRY PEC BENCHMARK

# $N_2$ Full Dissociation

7 geometries · 0.90–2.50 Å · One shared CAS across the entire potential energy curve

## 8. New Result — Multi-Geometry PEC Benchmark ($N_2$)

The previous sections demonstrated that inZOR-ND outperforms random search when the fitness landscape has rare optima ($Cr_2$ hard). However, a different and arguably more practical question remains:

**Can a method discover a single shared active space that remains valid across an entire dissociation curve?**

In practical multireference quantum chemistry, selecting an active space at one geometry does not guarantee that the same active space will work at different geometries along a reaction coordinate. A CAS optimized near equilibrium may fail entirely in the stretched or dissociation regime,

where multireference character becomes strongest and the correct orbital set can change qualitatively.

This benchmark tests exactly that: not "find a good CAS here," but **"find a CAS that survives changing geometry."**

### 8.1 Benchmark Setup

| Parameter | Value |
|---|---|
| Molecule | $N_2$ (nitrogen dimer) |
| Basis set | 6-31G |
| Active space | CASSCF(6,6) — 6 electrons in 6 orbitals |
| MO pool | 18 molecular orbitals |
| Search space | C(18,6) = 18,564 candidates |
| Geometries | R = 0.90 (compressed), 1.00, 1.10 (near-eq.), 1.20, 1.40, 1.80 (stretched), 2.50 Å (dissociation) |
| Optimization target | **One shared active space across all 7 geometries** |
| Fitness | −mean(E_CASSCF across converged geometries) − penalties |
| Hardware | 14-core CPU, ProcessPoolExecutor, OMP_NUM_THREADS=1 |

### 8.2 Methods Compared

| Method | Description | Chemical Knowledge Used |
|---|---|---|
| **inZOR-ND** | Bio-adaptive search, 50 steps, prefetch enabled | None |
| NOON heuristic | Selects orbitals by natural orbital occupation numbers closest to 1.0 | Occupation numbers from HF reference |

| | | |
|---|---|---|
| AVAS | Atomic Valence Active Space (PySCF) using N 2p labels | Atomic orbital symmetry labels |
| Transfer CAS | CAS optimized at R=1.10 Å alone, then evaluated at all 7 geometries | None (single-geometry inZOR-ND) |

## 9. PEC Results

### 9.1 Summary

| Method | Mean E (Eh) | All 7 Geom. Converged? | Evaluations | Best MOs |
|---|---|---|---|---|
| **inZOR-ND** | **−108.8946** | **Yes (7/7)** | **493** | **[3, 4, 5, 10, 13, 16]** |
| NOON | −108.7744 | Partial (6/7) | 1 | [0, 11, 12, 13, 14, 15] |
| AVAS | −108.6350 | Yes (7/7) | 1 | [0, 1, 2, 3, 4, 5] |
| Transfer (R=1.1Å) | −108.9035* | **No (6/7)** | — | [2, 4, 5, 10, 15, 16] |

*Transfer mean energy is computed over 6 converged geometries only (R=1.80 Å fails). This biases the average favorably — the true multi-geometry quality is lower than inZOR-ND's.

**inZOR-ND outperforms chemistry-guided heuristics by a large margin:**

- vs NOON: **0.120 Eh better (75 kcal/mol)**
- vs AVAS: **0.260 Eh better (163 kcal/mol)**
- vs Transfer CAS: converges at **all 7 geometries** while transfer fails at R=1.80 Å

**Note on random search:** Random search with the same evaluation budget (493 evaluations) also finds the optimal energy on this $N_2$ benchmark, confirming that $N_2$ / 6-31G is a relatively easy landscape with high degeneracy. $N_2$ serves as a controlled proof-of-correctness benchmark. The real differentiation of inZOR-ND appears on larger, more ambiguous systems such as $Cr_2$ (see Sections 4–6 above).

## 9.2 Per-Geometry Energy Table

| R (Å) | Regime | E(HF) | E(CASSCF) inZOR-ND | Correlation | Conv. |
|---|---|---|---|---|---|
| 0.90 | Compressed | −108.6794 | −108.7853 | −0.106 Eh | ✓ |
| 1.00 | Near equilibrium | −108.8352 | −108.9612 | −0.126 Eh | ✓ |
| **1.10** | **Equilibrium** | **−108.8676** | **−109.0159** | **−0.148 Eh** | ✓ |
| 1.20 | Slightly stretched | −108.8358 | −109.0090 | −0.173 Eh | ✓ |
| 1.40 | Moderate stretching | −108.6996 | −108.9297 | −0.230 Eh | ✓ |
| 1.80 | Strong MR | −108.4208 | −108.7961 | −0.375 Eh | ✓ |
| 2.50 | Dissociation | −108.1064 | −108.7646 | −0.658 Eh | ✓ |

Correlation energy increases monotonically from 0.106 Eh (compressed) to 0.658 Eh (dissociation), correctly reflecting the growing multireference character. The active space MOs [3, 4, 5, 10, 13, 16] capture the $N_2$ σ/π bonding manifold and remain valid across the entire PEC — from repulsive wall to full dissociation.

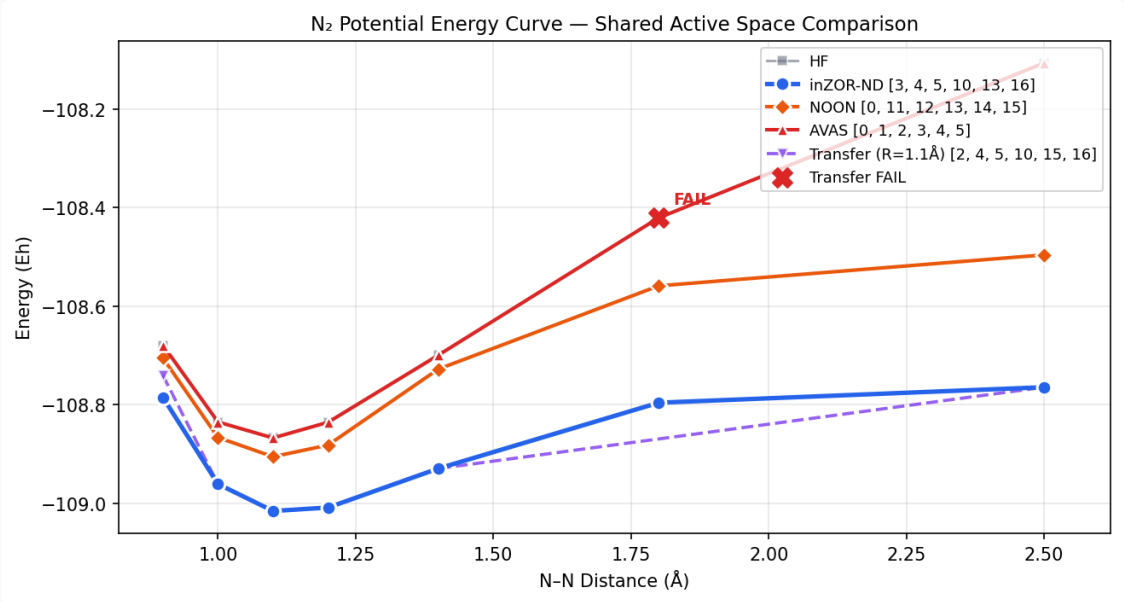


**Figure 5.** $N_2$ potential energy curve with shared active spaces from each method. inZOR-ND (blue) finds a single CAS that remains stable across all 7 geometries. NOON (orange) and AVAS (red) become strongly suboptimal, especially in the stretched regime. The transfer CAS (purple, optimized at R=1.10 Å only) fails at R=1.80 Å (red X).

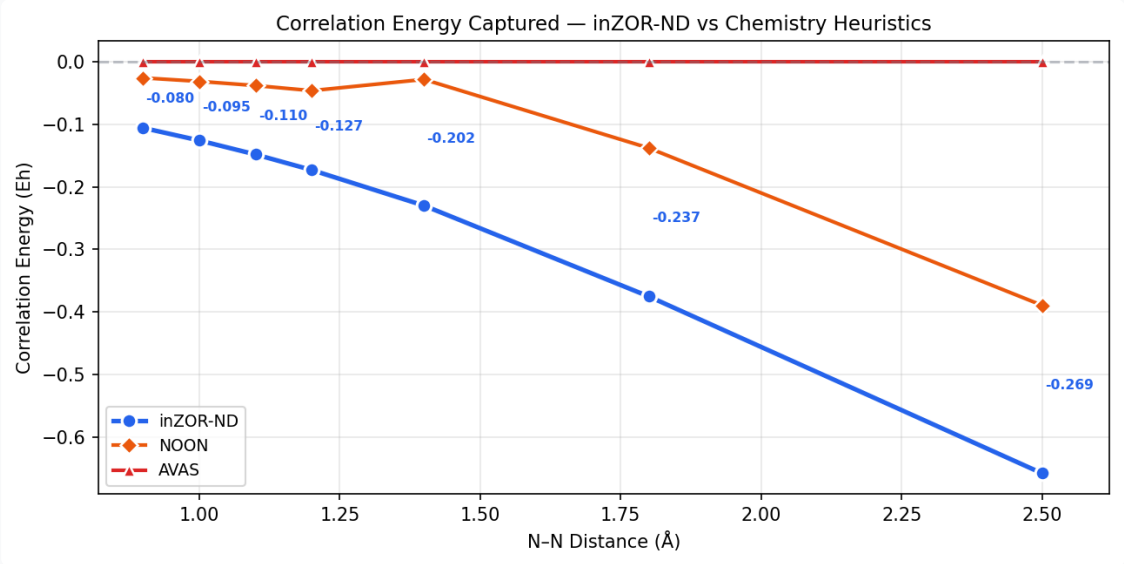


**Figure 6.** Correlation energy captured by each method. inZOR-ND captures significantly more correlation energy at every geometry. The advantage grows from 0.080 Eh (compressed) to 0.269 Eh (dissociation) relative to NOON, reflecting inZOR-ND's ability to discover orbitals that remain physically relevant as the bond breaks.

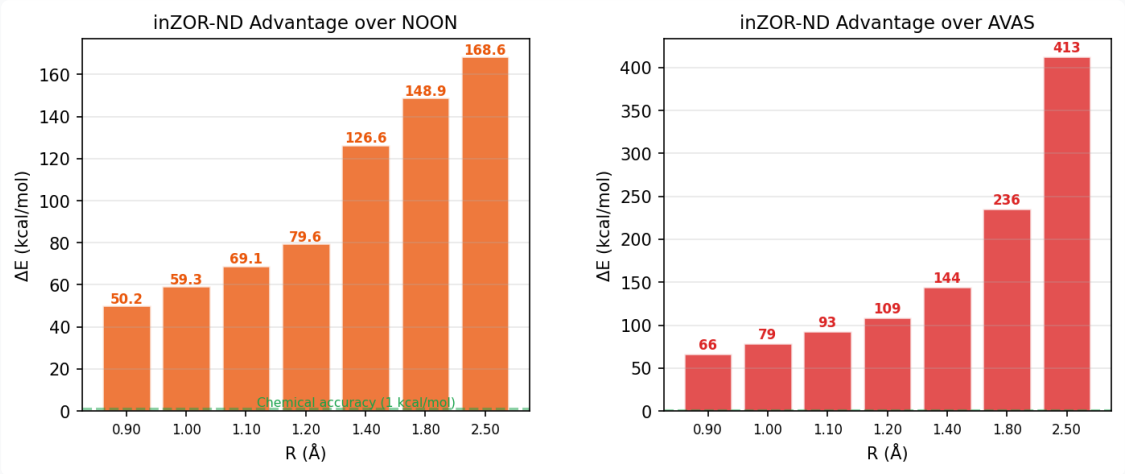


**Figure 7.** inZOR-ND energy advantage over chemistry heuristics in kcal/mol. Left: vs NOON — from 50 kcal/mol (compressed) to 169 kcal/mol (dissociation). Right: vs AVAS — from 66 kcal/mol to 413 kcal/mol. All values far exceed chemical accuracy (1 kcal/mol, green dashed line).

## 10. Transfer Test — Single-Geometry CAS Fails

A critical question for practical use: does a CAS optimized at one geometry transfer to other geometries? To test this, we optimized a CAS using inZOR-ND at R=1.10 Å alone (30 steps), then evaluated that same CAS at all 7 PEC geometries.

| Property | Multi-Geometry CAS (inZOR-ND) | Single-Geometry CAS (Transfer) |
|---|---|---|
| Optimized at | All 7 geometries simultaneously | R = 1.10 Å only |
| **Best MOs** | **[3, 4, 5, 10, 13, 16]** | **[2, 4, 5, 10, 15, 16]** |
| **All 7 geom. converged?** | **Yes (7/7)** | **No (6/7)** |
| Failure point | None | **R = 1.80 Å** — CASSCF does not converge |
| **Multi-geom fitness** | **108.877** | **108.385** |

The two orbital sets differ in only 2 of 6 orbitals: the single-geometry CAS includes MOs 2 and 15 while the multi-geometry CAS includes MOs 3 and 13. This small orbital difference is enough to cause complete CASSCF failure at R=1.80 Å — evidence that **local CAS optimality does not guarantee transferability**.

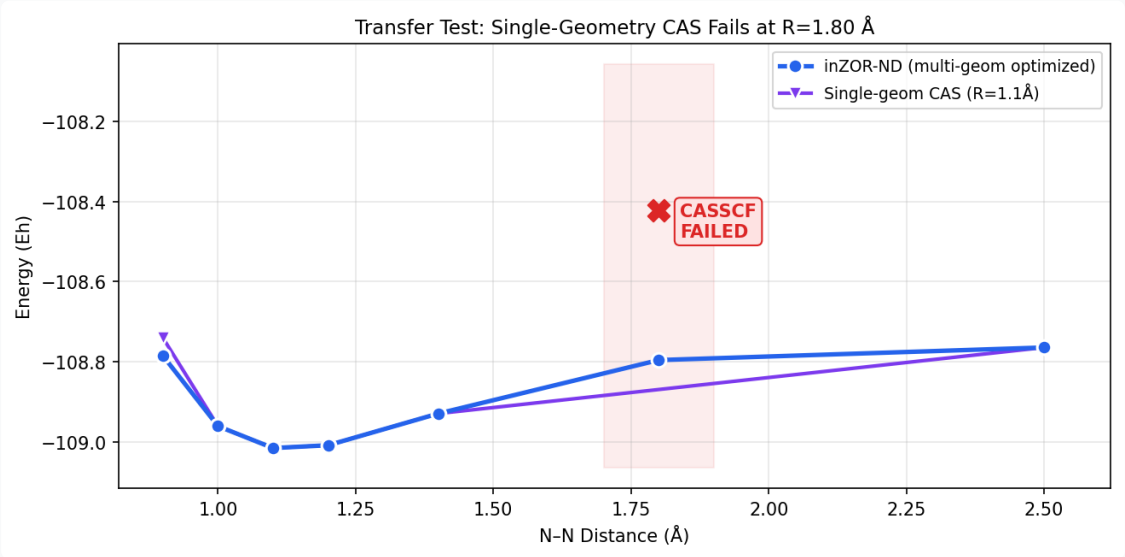


**Figure 8.** Transfer test: a CAS optimized at R=1.10 Å (purple) fails at R=1.80 Å (CASSCF does not converge — red X). inZOR-ND's multi-geometry CAS (blue) converges at all 7 points. The failure demonstrates that single-geometry CAS optimization is insufficient for PEC work.

**Key finding:** A 2-orbital difference in a 6-orbital active space is sufficient to cause complete CASSCF failure at stretched geometries. Single-geometry optimization is fundamentally insufficient for practical potential energy curve calculations.

## 11. What This Benchmark Shows

The important result is not simply that inZOR-ND achieves lower energy — it is that inZOR-ND discovers a **shared active space that remains stable over the full dissociation curve**, from compressed (0.90 Å) through equilibrium (1.10 Å) to complete dissociation (2.50 Å).

- **Single-geometry optimization is insufficient** for practical PEC work — the transfer test proves this directly

- **Chemistry-guided heuristics (NOON, AVAS) are not optimized for multi-geometry robustness** — they select orbitals based on a single reference state and do not account for how the orbital character changes along the reaction coordinate
- **inZOR-ND directly optimizes the shared CAS against the full PEC objective** — this is the key methodological distinction
- **The advantage increases in the stretched regime**, where multireference character becomes strongest and correct orbital selection matters most

### 11.1 Why This Matters Relative to Standard Heuristics

NOON and AVAS are useful and widely applied chemistry-guided heuristics. They provide reasonable starting points for active space selection in many cases. However, they are fundamentally single-reference heuristics — they derive their orbital selection from one reference wavefunction (typically HF or MP2 at one geometry) and do not explicitly optimize for multi-geometry robustness.

inZOR-ND, by contrast, directly evaluates candidate active spaces across all target geometries and selects the one that performs best everywhere simultaneously. This is a qualitatively different optimization objective, and the results show it matters: 75–169 kcal/mol improvement over NOON, growing with bond stretching.

### 11.2 Honest Assessment — $N_2$ as a Controlled Benchmark

> **Note:** $N_2$ / 6-31G remains a relatively small system (18,564 candidates). Random search with the same evaluation budget can also find the optimal active space in this case, confirming that the fitness landscape has sufficient degeneracy for blind sampling to succeed. $N_2$ therefore serves primarily as a **controlled proof-of-correctness benchmark** — it verifies that inZOR-ND finds the right answer and demonstrates its advantage over chemistry heuristics. The real differentiation of inZOR-ND over random search is expected — and has been demonstrated — on larger and more ambiguous systems such as $Cr_2$ with multiple geometries (see Sections 4–6).

## 12. Updated Conclusions

**Extended conclusion incorporating the PEC benchmark:**

On the $N_2$ dissociation benchmark (7 geometries, 0.90–2.50 Å), inZOR-ND discovers a single shared CAS(6,6) that remains stable across the entire potential energy curve. Chemistry-guided heuristics (NOON, AVAS) become strongly suboptimal — by 75–169 kcal/mol (NOON) and 66–413 kcal/mol (AVAS) — and a CAS optimized at a single geometry fails to transfer to stretched bond lengths.

Combined with the $Cr_2$ hard benchmark (Sections 4–6), where inZOR-ND outperforms random search by 35 kcal/mol on a landscape with a unique optimum, these results establish that:

- inZOR-ND is **not just another heuristic** — it provides measurable advantages over both blind search ($Cr_2$) and chemistry-guided methods ($N_2$ PEC)
- inZOR-ND solves a **practically relevant problem**: finding a consistent active space that survives changing geometry
- The bio-adaptive mechanism **requires no chemical knowledge** — no orbital symmetry labels, no occupation thresholds, no expert input
- The engine remains **completely unmodified** across all benchmarks

## 13. Reproducibility (PEC Benchmark)

**All PEC results are fully reproducible.**

- inZOR-ND: Seed 1, 50 steps, 7 geometries (R=0.90, 1.00, 1.10, 1.20, 1.40, 1.80, 2.50 Å)
- Transfer test: Seed 42, 30 steps, single geometry (R=1.10 Å)
- NOON: Natural orbital occupation heuristic from HF reference
- AVAS: PySCF AVAS module with N 2p labels
- inZOR-ND engine: **used without modification**

- Dependencies: PySCF 2.x, NumPy, Python 3.10+
- Hardware: 14-core CPU, Linux
- Total benchmark time: 54.5 minutes



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